Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Machado Dairy Farms #4

Physical address of dairy:

Number and Street	City	County	Zip Code
26298 Groves RD	Farmington	San Joaquin	95230

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1980

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X187-X190-XX03-XXXX X187-X300-XX14-XXXX

B. OPERATORS

Machado, John

Operator name: Machado, John

Telephone no.: Landline: ; Cellular: (209) 602-8598

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 4430	Mantecca	CA	95337

This operator is responsible for paying permit fees.

C. OWNERS

Amaral, Armelim

Legal owner name: Amaral, Armelim

Telephone no.: Landline: (209) 848-9502; Cellular:

Mailing Address Number and Street	City	State	Zip Code
P.O. Box 263	Farmington	CA	95230

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	0	0	0	0
Number under roof	715	1	0	0	0	0
Maximum number	720	1	0	0	0	0
Average number	715	1	0	0	0	0
Avg live weight (lbs)	1,300	1,300	0	0		

Predominant milk cow breed: Jersey-Holstein Cross

Average milk production: 75 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 18,776.96 *tons per reporting period*

Total nitrogen from manure: 245,464.97 *lbs per reporting period*

After ammonia losses (30% loss applied): 171,825.48 *lbs per reporting period*

Total phosphorus from manure: 41,628.91 *lbs per reporting period*

Total potassium from manure: 139,277.87 *lbs per reporting period*

Total salt from manure: 336,887.70 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 8,235,000 *gallons*

Total nitrogen generated: 50,157.37 *lbs*

Total phosphorus generated: 4,545.73 *lbs*

Total potassium generated: 80,943.66 *lbs*

Total salt generated: 558,225.42 *lbs*

	8,235,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	8,235,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Creek	Surface water
IW #1	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

Date	Material type / Description	Quantity	Reporting basis	Moisture (%)	N (%)	P (%)	K (%)	Salt (%)
06/24/2023	Solid commercial fertilizer *4-0-0*	12.10 *ton*	Dry-weight	0.1	4.000000	0.000000	0.000000	0.000000
07/15/2023	Solid commercial fertilizer *UN32*	0.55 *ton*	Dry-weight	0.1	32.000000	0.000000	0.000000	0.000000

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Commercial fertilizer / Other	1,318.68	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Total imports for all materials	1,318.68	0.00	0.00	0.00

G. NUTRIENT EXPORTS

Date	Material type	Quantity	Reporting basis	Moisture (%)	Density (lbs/cu ft)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/07/2023	Corral solids	1,200.00 *ton*	Dry-weight	46.8		17,000.00	8,200.00	68,700.00		32.20
10/22/2023	Corral solids	900.00 *ton*	Dry-weight	58.5		28,500.00	1,000.00	29,700.00		34.00

No liquid nutrient exports entered.

Material type	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Dry manure	42,995.10	11,216.76	109,902.06	665,109.60
Process wastewater	0.00	0.00	0.00	0.00
Total exports for all materials	42,995.10	11,216.76	109,902.06	665,109.60

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	111	111	2	process wastewater	X187-X300-XX14-XXXX
2	10	10	2	manure	X187-X300-XX14-XXXX
Heifer Pasture	28	28	0	none	X187-X190-XX03-XXXX
Hill Pasture	15	15	0	none	X187-X300-XX14-XXXX
Totals for areas that were used for application	121	121	4		
Totals for areas that were not used for application	43	43	0		
Land application area totals	164	164	4		

B. CROPS AND HARVESTS

1

Field name: 1

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 111 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/25/2023	1,816.00 *ton*	As-is		60.6	4,180.00	910.00	5,670.00		9.33

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.36	136.77	29.78	185.53	1,202.82

06/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 111 Plant date: 06/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	3,170.30 *ton*	As-is		61.0	3,820.00	530.00	5,270.00		7.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	28.56	218.21	30.27	301.04	1,604.00

2

Field name: 2

11/02/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 10 Plant date: 11/02/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/25/2023	161.00 *ton*	As-is		57.8	4,390.00	9,100.00	5,060.00		7.56

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.10	141.36	293.02	162.93	1,027.28

06/25/2023: Corn, silage

Crop: Corn, silage Acres planted: 10 Plant date: 06/25/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/23/2023	278.20 *ton*	As-is		60.4	4,400.00	640.00	5,160.00		7.10

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	28.00	224.00	42.00	184.80	0.00
Total actual harvest content	27.82	244.82	35.61	287.10	1,564.37

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 11/02/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
02/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	82.51	4.27	48.41	664.98	1,260,000.00 *gal*
Application event totals		82.51	4.27	48.41	664.98	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	99.46	12.90	176.19	1,366.91	1,890,000.00 *gal*
Application event totals		99.46	12.90	176.19	1,366.91	

1 - 06/25/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 06/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	3.09	0.00	0.00	0.00	14,208,000.00 *gal*
Application event totals		3.09	0.00	0.00	0.00	

1 - 06/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	138.53	12.20	259.01	1,538.37	2,610,000.00 *gal*
Canal	Surface water	2.42	0.00	0.00	0.00	11,136,000.00 *gal*
Application event totals		140.95	12.20	259.01	1,538.37	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	3.00	0.00	0.00	0.00	13,824,000.00 *gal*
Application event totals		3.00	0.00	0.00	0.00	

08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.96	0.00	0.00	0.00	13,632,000.00 *gal*
Application event totals		2.96	0.00	0.00	0.00	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
WWS	Process wastewater	131.37	11.57	245.61	1,458.80	2,475,000.00 *gal*
Canal	Surface water	2.29	0.00	0.00	0.00	10,560,000.00 *gal*
Application event totals		133.66	11.57	245.61	1,458.80	

09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.88	0.00	0.00	0.00	13,248,000.00 *gal*
Application event totals		2.88	0.00	0.00	0.00	

09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	2.88	0.00	0.00	0.00	13,248,000.00 *gal*
Application event totals		2.88	0.00	0.00	0.00	

2 - 11/02/2022: Oats, silage-soft dough

Field name: 2

Crop: Oats, silage-soft dough

Plant date: 11/02/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	178.21	137.74	156.20	3,266.00	100.00 *ton*
Application event totals		178.21	137.74	156.20	3,266.00	

2 - 06/25/2023: Corn, silage

Field name: 2

Crop: Corn, silage

Plant date: 06/25/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/07/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Corral Solids	Corral solids	253.23	122.15	1,023.36	4,796.51	140.00 *ton*
Application event totals		253.23	122.15	1,023.36	4,796.51	

06/24/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
4-0-0	Solid commercial fertilizer	8.00	0.00	0.00	0.00	
Application event totals		8.00	0.00	0.00	0.00	

06/26/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	3.34	0.00	0.00	0.00	1,302,000.00 *gal*
Application event totals		3.34	0.00	0.00	0.00	

2 - 06/25/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Sidedress	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
UN32	Solid commercial fertilizer	35.20	0.00	0.00	0.00	
Application event totals		35.20	0.00	0.00	0.00	

07/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	3.12	0.00	0.00	0.00	1,218,000.00 *gal*
Application event totals		3.12	0.00	0.00	0.00	

07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	3.23	0.00	0.00	0.00	1,260,000.00 *gal*
Application event totals		3.23	0.00	0.00	0.00	

08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	3.12	0.00	0.00	0.00	1,218,000.00 *gal*
Application event totals		3.12	0.00	0.00	0.00	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	3.12	0.00	0.00	0.00	1,218,000.00 *gal*
Application event totals		3.12	0.00	0.00	0.00	

09/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	3.12	0.00	0.00	0.00	1,218,000.00 *gal*
Application event totals		3.12	0.00	0.00	0.00	

2 - 06/25/2023: Corn, silage

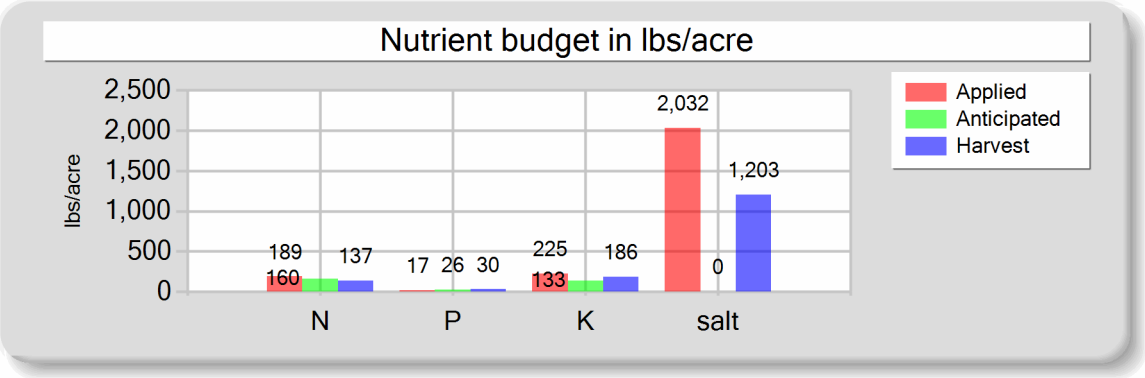
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Creek	Surface water	3.01	0.00	0.00	0.00	1,176,000.00 *gal*
Application event totals		3.01	0.00	0.00	0.00	

B. NUTRIENT BUDGET

1 - 11/02/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	181.97	17.17	224.60	2,031.89
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	188.97	17.17	224.60	2,031.89
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	136.77	29.78	185.53	1,202.82
Nutrient balance	52.20	-12.60	39.07	829.07
Applied to removed ratio	1.38	0.58	1.21	1.69

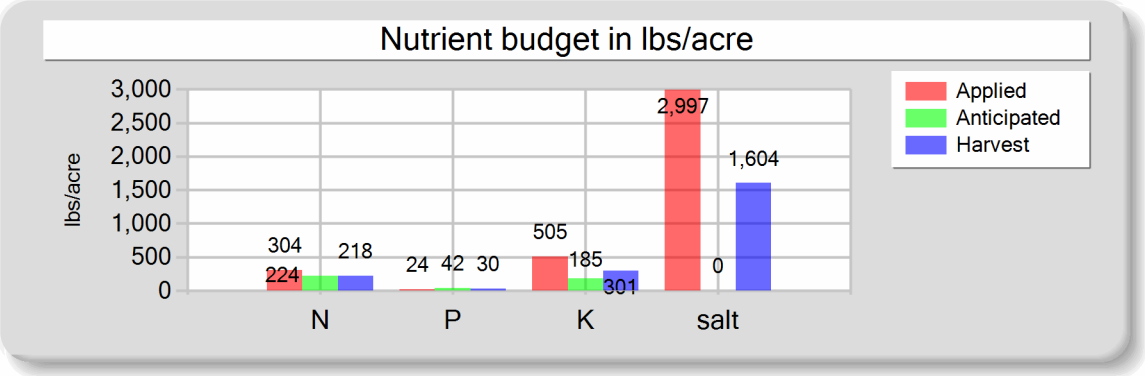
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
3,150,000.00 *gallons*
116.00 *acre-inches*
1.05 *inches/acre*

Total harvests for the crop
1 *harvests*

1 - 06/25/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	8.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	269.90	23.78	504.62	2,997.16
Fresh water	19.52	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	304.42	23.78	504.62	2,997.16
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	218.21	30.27	301.04	1,604.00
Nutrient balance	86.21	-6.50	203.59	1,393.16
Applied to removed ratio	1.40	0.79	1.68	1.87

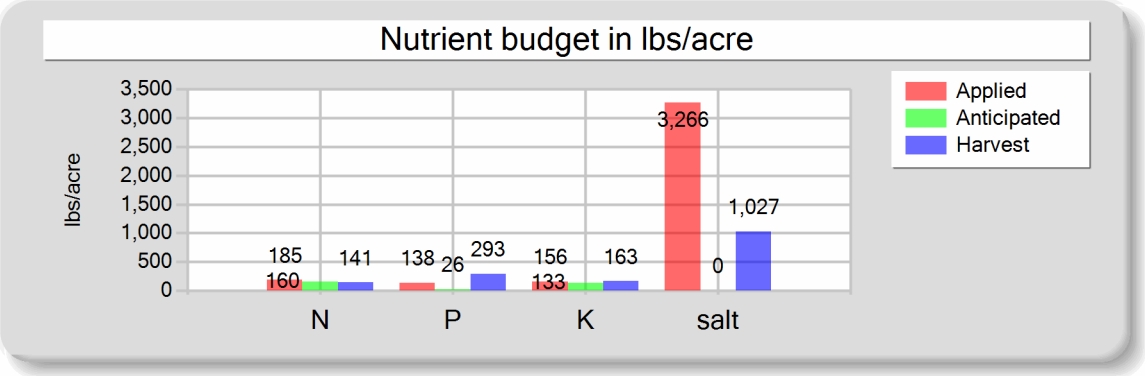
Fresh water applied	
89,856,000.00	*gallons*
3,309.09	*acre-inches*
29.81	*inches/acre*

Process wastewater applied	
5,085,000.00	*gallons*
187.26	*acre-inches*
1.69	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 11/02/2022: Oats, silage-soft dough

Field name: 2 Crop: Oats, silage-soft dough Plant date: 11/02/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	178.21	137.74	156.20	3,266.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	185.21	137.74	156.20	3,266.00
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	141.36	293.02	162.93	1,027.28
Nutrient balance	43.85	-155.28	-6.73	2,238.72
Applied to removed ratio	1.31	0.47	0.96	3.18

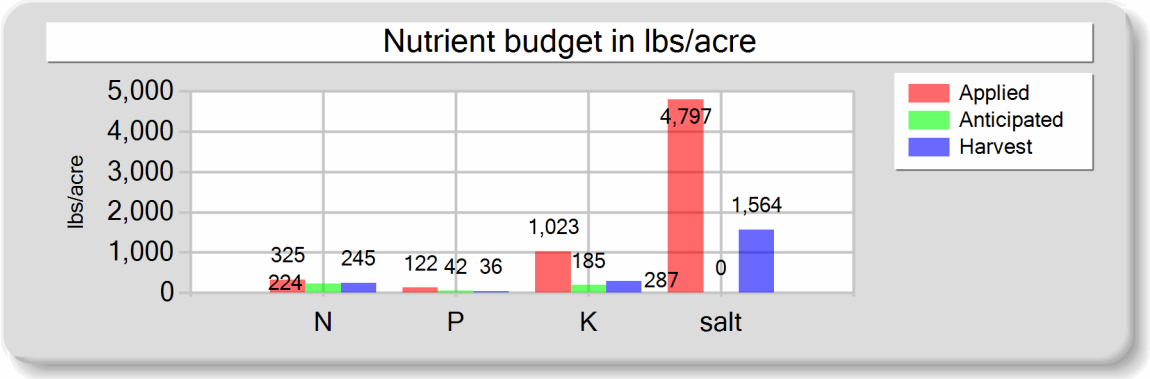
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
1 *harvests*

2 - 06/25/2023: Corn, silage

Field name: 2 Crop: Corn, silage Plant date: 06/25/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	43.20	0.00	0.00	0.00
Dry manure	253.23	122.15	1,023.36	4,796.51
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	22.06	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	325.49	122.15	1,023.36	4,796.51
Anticipated crop nutrient removal	224.00	42.00	184.80	0.00
Actual crop nutrient removal	244.82	35.61	287.10	1,564.37
Nutrient balance	80.67	86.54	736.25	3,232.14
Applied to removed ratio	1.33	3.43	3.56	3.07

Fresh water applied	
8,610,000.00	*gallons*
317.08	*acre-inches*
31.71	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

Corral Solids

Sample and source description: Corral Solids

Sample date: 09/12/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 64.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	25,100.00	19,400.00	22,000.00	54,900.00	18,800.00	5,290.00	8,700.00	3,440.00		46.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 04/17/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 46.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	17,000.00	8,200.00	68,700.00	14,400.00	8,300.00	7,300.00	4,600.00	4,340.00		32.20
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Compost

Sample and source description: Compost

Sample date: 10/09/2023 Material type: Compost Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 22.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	20,800.00	9,400.00	10,000.00	0.00	0.00	0.00	0.00	0.00		24.20
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

Corral Solids

Sample and source description: Corral Solids

Sample date: 10/09/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 58.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	28,500.00	1,000.00	29,700.00	0.00	0.00	0.00	0.00	0.00		34.00
DL	100.00	100.00	100.00	100.00	100.00	100.00	100.00	50.00		0.01

B. PROCESS WASTEWATER ANALYSES

WWQ1

Sample and source description: WWQ1

Sample date: 02/21/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	871.00	543.00	0.00	0.00	45.10	511.00	109.00	61.50	170.00	4,680.00	0.00	617.00	912.00	10,000.00	7,020
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	100.00	10

WWQ2

Sample and source description: WWQ2

Sample date: 04/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	700.00	521.00	0.00	0.00	90.80	1,240.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	9,650.00	9,620
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ3

Sample and source description: WWQ3

Sample date: 07/26/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	706.00	568.00	0.00	0.00	62.20	1,320.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	10,100.00	7,840
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

WWQ4

Sample and source description: WWQ4

Sample date: 10/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	647.00	423.00	0.00	0.00	60.00	1,260.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	8,550.00	9,100
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

C. FRESH WATER ANALYSES

Canal

Canal

Sample description: Canal

Sample date: 11/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.89	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

Creek

Creek

Creek

Sample description: Creek

Sample date: 11/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	3.07	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

IW #1

IW #1

Sample description: IW #1

Sample date: 11/02/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.91	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0
DL	0.50	0.50	0.50	0.50	0.50	0.50	0.50	2.00	1.00	0.10	1.00	10

D. SOIL ANALYSES

1

Soil

Sample and source description: Soil

Sample date: 10/23/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			112.00				
DL			4.00				

2

Soil

Sample and source description: Soil

Sample date: 10/23/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			123.00				
DL			4.00				

E. PLANT TISSUE ANALYSES

1 - 11/02/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,180.00	910.00	5,670.00		9.33
DL	100.00	100.00	100.00		0.01

1 - 06/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 61.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	3,820.00	530.00	5,270.00		7.20
DL	100.00	100.00	100.00		0.01

2 - 11/02/2022: Oats, silage-soft dough

Forage

Sample and source description: Forage

Sample date: 05/30/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 57.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,390.00	9,100.00	5,060.00		7.56
DL	100.00	100.00	100.00		0.01

2 - 06/25/2023: Corn, silage

Corn Silage

Sample and source description: Corn Silage

Sample date: 10/23/2023 Source of analysis: Lab analysis Method of reporting: As-is

Moisture: 60.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	4,400.00	640.00	5,160.00		7.10
DL	100.00	100.00	100.00		0.01

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

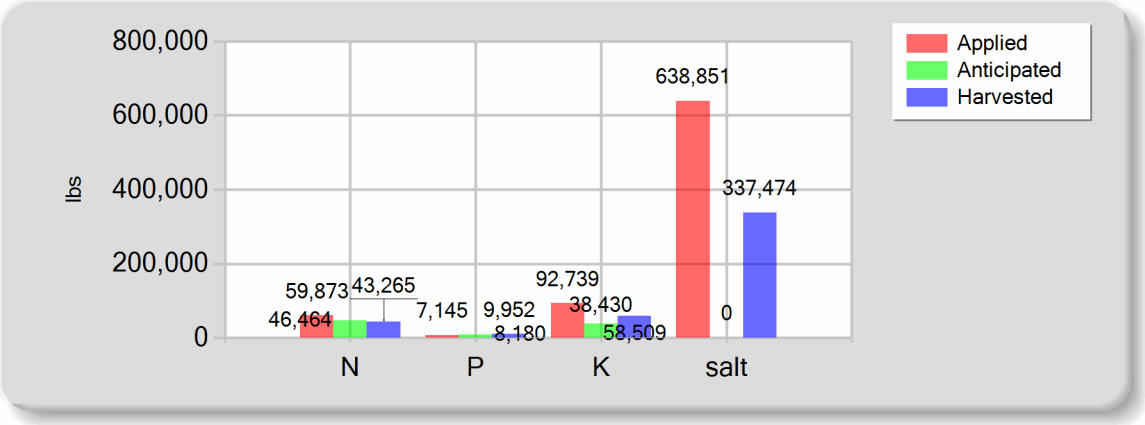
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

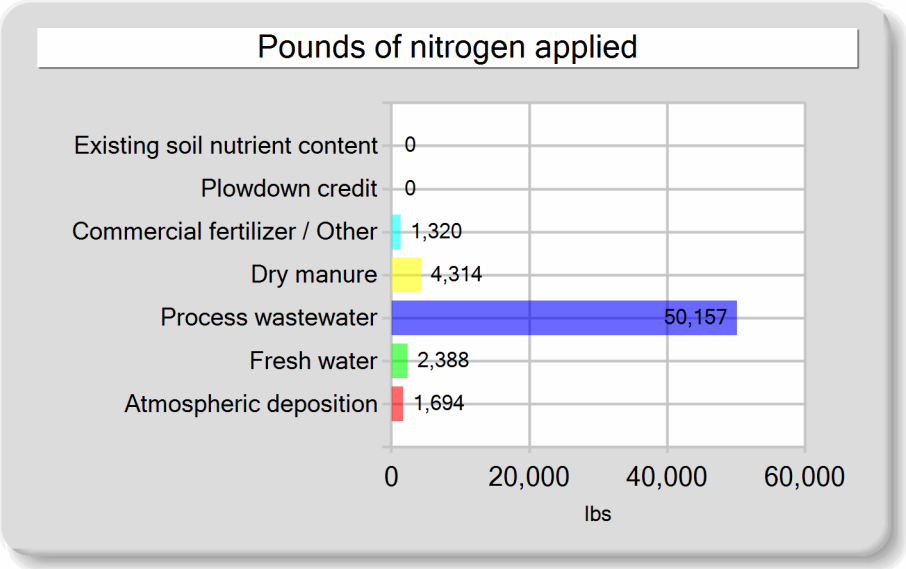
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

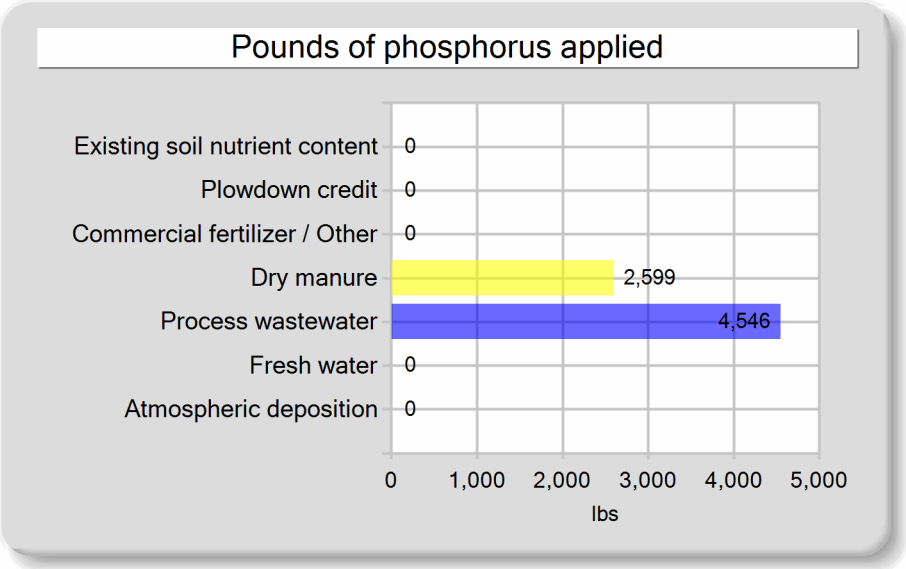
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	1,320.00	0.00	0.00	0.00
Dry manure	4,314.42	2,598.87	11,795.55	80,625.12
Process wastewater	50,157.37	4,545.73	80,943.66	558,225.42
Fresh water	2,387.64	0.00	0.00	0.00
Atmospheric deposition	1,694.00	0.00	0.00	0.00
Total nutrients applied	59,873.43	7,144.60	92,739.21	638,850.54
Anticipated crop nutrient removal	46,464.00	8,179.60	38,429.60	0.00
Actual crop nutrient removal	43,264.59	9,951.93	58,508.75	337,473.67
Nutrient balance	16,608.84	-2,807.33	34,230.47	301,376.88
Applied to removed ratio	1.38	0.72	1.59	1.89

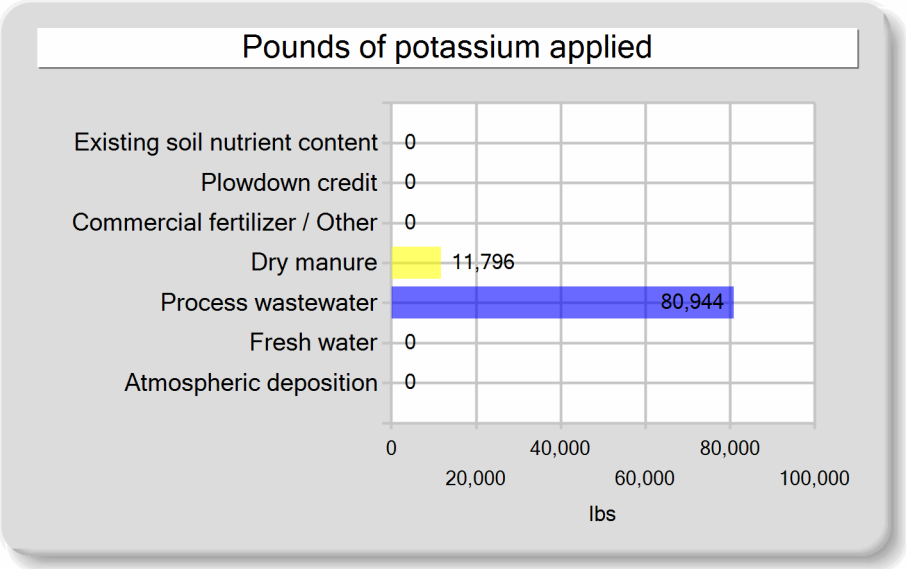
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

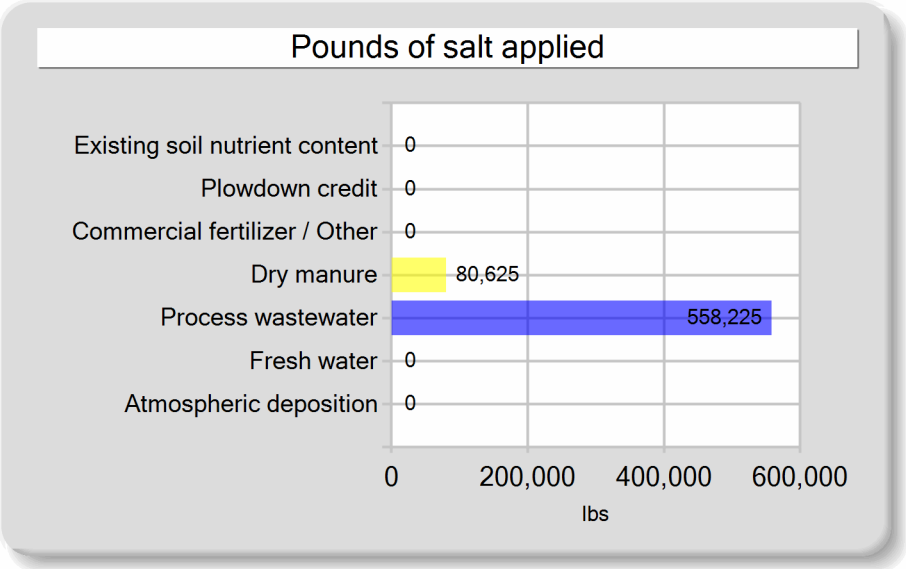


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

Armelim Amaral	John A Machado John Machado (Apr 12, 2024 07:06 PDT)
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Armelim Amaral	John Machado
PRINT OR TYPE NAME	PRINT OR TYPE NAME
4/8/24	Apr 12, 2024
DATE	DATE

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: John Machado

Name of Dairy Facility: Machado Dairy Farms #4

Facility Address: 26298 Groves Road, Farmington, CA 95230

Number and Street | City | Zip Code

Contact Person Name and Phone Number: John Machado 209-602-8598

Name | Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Machado Dairy Farms

Address of Hauling Company /Person: PO Box 4430 Manteca, CA 95337

Number and Street | City | Zip Code

Contact Person: John Machado 209-602-8598

Name | Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Tony Borba Dairy 23844 Dodds Road, Escalon, CA 95320 209-604-1764

Name | Number and Street | City | Zip Code | Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

23844 Dodds Road, Escalon, CA 95320 207-170-02, 207-180-01

Number and Street | City | Zip Code | Assessor's Parcel Number

Dates Hauled: 10/22/2023

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 900 Tons / Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 58.5% moisture

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

45 loads at 20 ton per load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: John Machado (Apr 12, 2024 07:06 PDT) Date: Apr 12, 2024

Hauler's Signature: John Machado (Apr 12, 2024 07:06 PDT) Date: Apr 12, 2024

ATTACHMENT D

Manure/Process Wastewater Tracking Manifest
For
Existing Milk Cow Dairies

Instructions:

1) Complete one manifest for each hauling event, for each destination. A hauling event may last for several days, as long as the manure is being hauled to the same destination.
2) If there are multiple destinations, **complete a separate form for each destination.**
3) The operator must obtain the signature of the hauler upon completion of each manure-hauling event.
4) The operator shall submit copies of manure/process wastewater tracking manifest(s) with the Annual Monitoring Report for Existing Milk Cow Dairies.

Operator Information:

Name of Operator: John Machado

Name of Dairy Facility: Machado Dairy Farms #4

Facility Address: 26298 Groves Road, Farmington, CA 95230

Number and Street | City | Zip Code

Contact Person Name and Phone Number: John Machado 209-602-8598

Name | Phone Number

Manure/Process Wastewater Hauler Information:

Name of Hauling Company/Person: Machado Dairy Farms

Address of Hauling Company /Person: PO Box 4430 Manteca, CA 95337

Number and Street | City | Zip Code

Contact Person: John Machado 209-602-8598

Name | Phone Number

Destination Information:

Composting Facility / Broker / Farmer / Other (identify) (please circle one)

Contact information of Composting Facility, Broker, Farmer, or Other (as identified above):

Tony Borba Dairy 23844 Dodds Road, Escalon, CA 95320 209-604-1764

Name | Number and Street | City | Zip Code | Phone Number

Manure/Process Wastewater Destination Address or Assessor's Parcel Number:

23844 Dodds Road, Escalon, CA 95320 207-170-02, 207-180-01

Number and Street | City | Zip Code | Assessor's Parcel Number

Dates Hauled: 5/7/2023

Amount Hauled:

Enter the amount of manure hauled in tons or cubic yards (indicate the units used), the manure solids content (if amount reported in tons) or manure density (if amount reported in cubic yards), and the method used to calculate the amount:

Manure: 1200 Tons / Cubic Yards (indicate which units used)

Manure Solids Content (if amount reported in tons): 46.8% moisture

Manure Density (if amount reported in cubic yards):

Method used to determine amount of manure: _____

60 loads at 20 ton per load

Enter the amount of process wastewater hauled in gallons and the method used to determine the amount.

Process Wastewater: _____ Gallons

Method used to determine volume of process wastewater: _____

_____

Written Agreement:
Does the Operator have a written agreement (in compliance with Land Application Specification C.2 of Waste Discharge Requirements General Order No. R5-2007-0035) with any party that receives process wastewater from the Operator for its own use? (please check one)

____ Yes _____ No

If the answer is no, the Operator agrees to have such a written agreement with any such party for any process wastewater transferred after **31 December 2007** to such party.
_____ (Operator shall provide initials here to acknowledge this requirement).

Certification:
I declare under the penalty of law that I personally examined and am familiar with the information submitted in this document, and that based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of a fine and imprisonment for knowing violations.

Operator's Signature: John Machado (Apr 12, 2024 07:06 PDT) Date: Apr 12, 2024

Hauler's Signature: John Machado (Apr 12, 2024 07:06 PDT) Date: Apr 12, 2024

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Machado Dairy Farms #1,2,3 & 4
P.O. Box 4430
Manteca, Ca 95337

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: #4
RECEIVED DATE: 11/2/23 11:01 AM
SUBMITTED BY: Jeanette
ANALYZED DATE: 11/2/23

Irrigation/ Ground Water (H1)
SOURCE: Well Water

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Milk Barn	T3306002A	Electrical Conductivity (EC)	EPA 120.1 and	0.150	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	0.303	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	95.9	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	95.6	97.6 (T3306002A)	95.2 (T3306002A)	2.48
Electrical Conductivity (EC)	98.9			0.297
Nitrate Nitrogen (NO3-N)	97.2	99.6 (T3305011A)	91.5 (T3305011A)	8.52

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

www.denelelabs.com

Compliance Analysis Report

Customer
Machado Dairy Farms #1,2,3 & 4
P.O. Box 4430
Manteca, Ca 95337

Grower

Consultant:
F & R Ag Services
2857 Geer Rd. Ste A
Turlock, CA 95382

PURCHASE ORDER: #4
RECEIVED DATE: 11/2/23 11:01 AM
SUBMITTED BY: Jeanette
ANALYZED DATE: 11/2/23

Irrigation / Ground Water (H7)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 7.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
IW 1	T3306002B	Electrical Conductivity (EC)	EPA 120.1 and	< 0.100	mmhos/cm
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	< 0.140	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	< 10.0	mg/L
		Total Nitrogen (TN)	SM 4500-NH3 D,C	2.91	mg/L
		Soluble Salts (SALT-SOL)	Calculation	16.6	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	95.6	97.6 (T3306002A)	95.2 (T3306002A)			2.48
Electrical Conductivity (EC)	98.9			4.05 (T3305038A)	4.04 (Dup)	0.297
Nitrate Nitrogen (NO3-N)	97.2	99.6 (T3305011A)	91.5 (T3305011A)			8.52
Total Dissolved Solids (TDS)	99.0			140 (T3310046G)	135 (Dup)	3.64
Total Nitrogen (TN)	91.3			2.80 (T3297037A)	2.80 (null)	0.000

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DENELE ANALYTICAL, INC.

865 S. KILROY ROAD
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: Machado # 4
Mailing address: 1,2,3,4
(not facility address)
Phone#:

Authorized Copy Released To: F&R AG SERVICES
Sampled By: Jeanette
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH3-N*, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	11/2/23		Milk Barn	Domestic	4.0°c		X				
2	↓		IW 1	Irrigation	7.2°c						H7
3			Canal	↓	6.4°						H6
4			Creek		7.0°c						H6
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

Signature	Company	Received Date & Time	Relinquished Date & Time	Logged in By:	LABORATORY NUMBER
Jeanette Sewell	F&R AG SERVICES				T3306002
2nd Maria Orles	DENELE ANALYTICAL, INC.	11-2-23 11:01 am			

(LABORATORY USE ONLY)

Machado Dairy Farms #4 2023 Annual Report

Final Audit Report 2024-04-12

Created:	2024-04-11
By:	Vince Furtado (vfurtado@fragservices.com)
Status:	Signed
Transaction ID:	CBJCHBCAABAAbQyuRTxiFsnphFykLka2NbE1QrO6N0lg

"Machado Dairy Farms #4 2023 Annual Report" History

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